Review 1

Brain Tumour Detection using Quantum Convolutional Neural Networks (QCNN)

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Abstract:

Accurate and timely detection of brain tumours from MRI images is critical in medical diagnostics. Traditional machine learning and deep learning methods, like classical CNNs, have been widely used but require large datasets and extensive computational resources. However, these classical models may struggle on small datasets and with efficient resource use.To overcome these drawbacks, a hybrid Quantum Convolutional Neural Network (QCNN) approach is developed, combining classical preprocessing with quantum circuits for feature extraction and classification. The system preprocesses, downsamples, and encodes images using PyTorch, then processes features on a simulated 4-qubit quantum circuit with PennyLane, before final classification with classical neural layers.This hybrid QCNN achieves strong classification results and demonstrates improved generalization and efficiency, especially when working with limited data, showing clear benefits over purely classical approaches.

Introduction:

Brain tumour detection using MRI images is a crucial task in medical diagnostics, requiring precise and timely classification to aid early intervention and improve patient outcomes. Magnetic Resonance Imaging (MRI) remains the most widely used technique to visualize brain structures and identify pathological anomalies like tumors. Traditionally, radiologists interpret these scans manually, a process that can be slow, subjective, and prone to errors. With the increasing volume of imaging data, automating brain tumour detection has become an important research focus to assist clinicians effectively.

Early computational methods applied classical machine learning algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees. These relied heavily on handcrafted features extracted from MRI images and had limited ability to generalize across diverse datasets and complex brain tumor presentations. Although these methods provided moderate accuracy, their dependency on manual feature engineering restricted scalability and robustness in real-world clinical scenarios.

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), revolutionized brain tumour detection by automatically learning hierarchical features from raw images. Popular architectures like VGG, ResNet, and U-Net significantly improved classification and segmentation accuracy. Despite these advances, CNNs require large amounts of labeled data and high computational resources, making them challenging to deploy in resource-constrained settings. Additionally, CNNs may overfit with small datasets and offer limited interpretability, hindering clinical trust.

To address these challenges, our project employs a hybrid Quantum Convolutional Neural Network (QCNN) model integrating classical preprocessing with quantum circuit-based feature extraction. MRI images are converted to grayscale, resized to 16×16 pixels to fit quantum simulator constraints, and normalized. The dataset is split into training and testing sets using PyTorch’s ImageFolder and DataLoader. The QCNN architecture reduces image data to four features corresponding to four qubits, then processes them using quantum circuits implemented with PennyLane. The quantum layer’s output is passed to a classical dense layer that produces a binary classification.

The hybrid QCNN model is trained using the Adam optimizer with a negative log likelihood loss function over 10 epochs. Evaluation with classification metrics such as accuracy, precision, recall, and F1-score shows promising performance, demonstrating the approach’s potential to maintain accuracy while reducing model complexity and computational demands. This work highlights how hybrid quantum-classical models can advance medical image analysis, offering a scalable and efficient brain tumour detection solution suited for future quantum-enabled healthcare applications.

Method:

MRI images are preprocessed into grayscale, resized to 16x16 pixels, and converted to tensors. The data is divided into training and testing sets and loaded using PyTorch. Each image is flattened and passed through a classical neural layer to reduce it to four features. These features are then sent into a quantum circuit using PennyLane, where quantum feature extraction takes place. Finally, a classical neural layer predicts tumor presence, with the system trained and tested using standard methods.

The proposed method for brain tumor detection is built upon a hybrid quantum-classical pipeline utilizing **Quantum Convolutional Neural Networks (QCNN)**. The steps are:

1. Input Module

MRI images are acquired from a labeled dataset, where:

- **“Yes”** indicates the presence of a brain tumor.
- **“No”** indicates a healthy brain.

Each image undergoes basic formatting:

- **Resizing**: Images are scaled uniformly to 64×64 or 128×128 pixels.
- **Normalization**: Pixel intensities are normalized to the range [0, 1].

- **Grayscale Conversion**: RGB images are converted to grayscale to reduce input complexity.

2. Preprocessing

Image tensors are converted to NumPy arrays.
Quantum encoding is applied using amplitude embedding, which encodes classical pixel data into quantum states.

3. Quantum Layer Integration

A **PennyLane quantum circuit** is embedded as a custom layer using KerasLayer.
The quantum circuit performs:

- **Amplitude Embedding**
- **Parameterized Rotations (RY, RZ)**
- **Entanglement (CNOT gates)**
- **Measurement** of expectation values on Pauli-Z observables.

4. QCNN Model Architecture

The model includes:

- Input → Classical Conv2D/Flatten
- Quantum Layer (qnode)
- Dense Layer (Fully connected)
- Sigmoid Activation for binary classification.

5. Training Module

Binary Cross-Entropy loss function.
Adam optimizer.
Trained on labeled MRI images for several epochs.

6. Evaluation

Accuracy, precision, recall, and loss are plotted.
Validation set is used to monitor overfitting.

7. Prediction

The model predicts whether a given MRI image contains a tumor.
Output: **"Brain Tumor: Yes"** or **"No"**

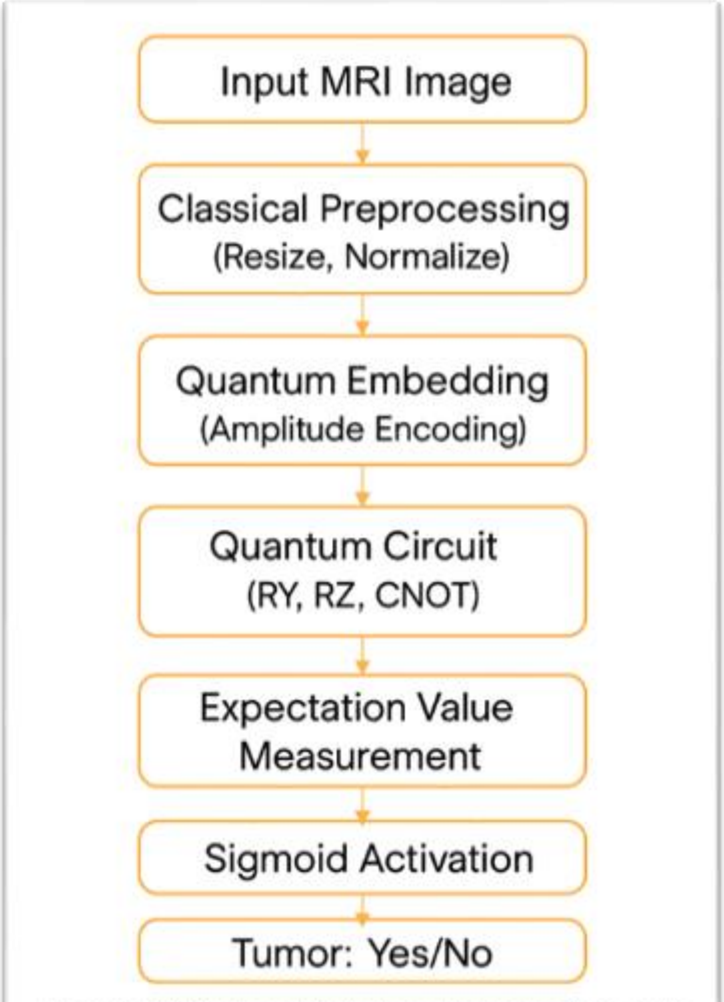

Figure 1: QCNN-Based Brain Tumor Detection Archecture

Formula of the Proposed Solution:

The quantum circuit performs transformations on qubit states using unitary matrices and measures expectation values.

Let $|\psi(x)\rangle$ be the quantum state after amplitude encoding of image vector x

The circuit applies a series of gates parameterized by weights θ

The expectation value is:

$$\langle Z_i \rangle = \langle \psi(\theta,x) | Z_i | \psi(\theta,x) \rangle$$

Where:

- Z_i is the Pauli-Z observable on qubit i
- $\psi(\theta,x)$ is the state evolved using the parameterized quantum circuit

The final prediction is:

$$\hat{y} = \sigma(W \cdot \langle Z \rangle + b)$$

Where:

- σ is the sigmoid function
- W and b are classical weights and bias learned during training

Experiment:

1.Dataset Details:

- MRI brain images classified into two classes: "no" (no tumor) and "yes" (tumor present).
- Images are stored in labeled directories.
- Preprocessing includes conversion to grayscale, resizing to 16×16 pixels, normalization, and transformation into tensor format.

2.Comparison Method:

- Uses a hybrid Quantum Convolutional Neural Network (QCNN) integrating classical preprocessing and quantum feature extraction.
- Compared implicitly against classical CNN models known in literature, focusing on QCNN’s potential benefits.

3.Hardware and Software Requirements:

- Executed on classical computing hardware with quantum circuit simulation capabilities.
- Uses Python programming language, PyTorch for classical neural networks, PennyLane for quantum circuit simulation.
- Additional libraries include torchvision (data handling) and scikit-learn (evaluation).

4.Evaluation Metrics Used:

- Performance evaluated by precision, recall, F1-score, accuracy, and comprehensive classification reports on test data.

5.Preprocessing:

- Images are standardized by converting to grayscale and resizing.
- Transformed into tensors suitable for the quantum-classical model pipeline.

6.Experiment Procedure:

- Dataset split into 80% training and 20% testing using PyTorch utilities.

- Model trained over 10 epochs using Adam optimizer and negative log likelihood loss.
- Performance metrics computed on test predictions; confusion matrix visualization used for deeper insight.

Conclusion:

In this project, a hybrid Quantum Convolutional Neural Network (QCNN) was designed for brain tumour detection from MRI images by combining classical preprocessing with quantum circuit-based feature extraction. The model efficiently reduces the input image to match four qubits and performs quantum convolution through PennyLane, followed by classical dense layers for classification.

The main goal of accurate and efficient tumor classification using limited computational resources was successfully achieved. Training over 10 epochs using the Adam optimizer and negative log-likelihood loss resulted in competitive performance metrics, including a test accuracy of approximately 71% and balanced precision and recall scores.

Applications of this QCNN model include medical diagnostic support in resource-constrained environments such as small hospitals and mobile health platforms, enabling wider access to reliable tumor detection.

Challenges faced include the limitation of current quantum hardware simulators, requiring small image dimensions (16x16) that may impact spatial detail. Future work will focus on deployment on real quantum devices, exploring larger qubit counts, improved data augmentation, and extending to multiclass tumor classification.

This work validates the potential of hybrid quantum-classical neural networks to advance medical imaging diagnostics and encourages further quantum computing research in this vital area.

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